

INDUSTRY PROBLEM SOLVING TASK

1. Autonomous Vehicle Vision System Selection

An automobile company is developing an autonomous vehicle detection system that must operate under the following conditions:

- If accuracy $< 90\%$ $\rightarrow$ system fails
- If latency > 70 ms $\rightarrow$ unacceptable
- The vehicle supports moderate computational resources

Methods available:

- Viola-Jones: 82% accuracy, 30 ms latency, low computation
- YOLO: 93% accuracy, 60 ms latency, moderate computation
- Deep Learning: 97% accuracy, 90 ms latency, high computation

Task:

Compare the three methods and evaluate which method satisfies all the constraints. Justify your decision using logical reasoning based on accuracy, latency, and computational requirements.

ANSWERS:

Problem Analysis

An autonomous vehicle requires a vision system that can accurately detect objects such as vehicles, pedestrians, traffic signs, and obstacles. The system must make decisions quickly because delays or incorrect detections can affect vehicle safety.

The system has the following requirements:

1. Accuracy must be 90% or above.
2. Latency must not exceed 70 ms.

3. The available hardware supports moderate computational requirements.

The available methods are compared below.

Comparison of Methods

Method	Accuracy	Latency	Computational Requirement	Evaluation
Viola-Jones	82%	30 ms	Low	Fast but insufficient accuracy
YOLO	93%	60 ms	Moderate	Meets all requirements
Deep Learning	97%	90 ms	High	Accurate but exceeds latency limit

Evaluation of Viola-Jones

Viola-Jones provides a latency of only 30 ms, making it the fastest method among the three. It also requires low computational resources, which makes it easy to deploy on systems with limited hardware.

However, its accuracy is only 82%, which is below the required minimum accuracy of 90%. This means that the system may fail to correctly detect important objects. In an autonomous vehicle, such missed or incorrect detections can create serious safety problems.

Therefore, although Viola-Jones satisfies the latency and computational requirements, it fails the accuracy requirement and cannot be selected.

Evaluation of YOLO

YOLO provides an accuracy of 93%, which is higher than the minimum required accuracy of 90%. Its latency is 60 ms, which is within the acceptable limit of 70 ms.

YOLO requires moderate computational resources, which matches the available hardware capability of the vehicle. Therefore, it does not create an excessive computational burden on the system.

YOLO can also perform object detection efficiently and is suitable for detecting multiple objects in real-time. This makes it appropriate for autonomous vehicle applications where quick and accurate decisions are necessary.

Therefore, YOLO satisfies:

- Accuracy requirement: $93\% \geq 90\%$ ✓
- Latency requirement: $60\text{ ms} \leq 70\text{ ms}$ ✓
- Computational requirement: Moderate = Available moderate resources ✓

Evaluation of Deep Learning

The Deep Learning method provides the highest accuracy of 97%, which is excellent for object detection. However, its latency is 90 ms, which is greater than the maximum acceptable limit of 70 ms.

In addition, it requires high computational resources. Since the vehicle only supports moderate computational capability, deploying this method may require expensive hardware upgrades.

The higher latency could delay important decisions such as braking, obstacle avoidance, or pedestrian detection. Therefore, despite its high accuracy, Deep Learning does not satisfy all the practical deployment requirements.

Final Decision

Based on logical evaluation, YOLO is the most suitable method.

It is the only method that satisfies all three constraints simultaneously:

- High accuracy
- Acceptable latency
- Moderate computational requirement

Conclusion

Therefore, YOLO should be selected for the autonomous vehicle vision system. Viola-Jones is fast but does not provide sufficient accuracy, while Deep Learning provides excellent accuracy but exceeds the latency limit and requires high computational resources. YOLO provides the most practical balance between safety, accuracy, speed, and hardware capability.



2. Smart Factory Product Inspection Selection

A manufacturing company wants to deploy a real-time computer vision system to detect defective products on a production line. The system must operate under the following conditions:

- If accuracy < 95% → defects may be missed
- If processing time > 50 ms → production is affected
- Hardware supports moderate computation

Methods available:

- Traditional Image Analysis: 91% accuracy, 25 ms, low computation
- YOLO: 96% accuracy, 45 ms, moderate computation
- Deep Learning: 98% accuracy, 75 ms, high computation

Task:

Compare the three methods and evaluate which method satisfies all the constraints. Justify

your decision using logical reasoning based on accuracy, processing time, and computational requirements.

ANSWERS:

Problem Analysis

In a smart factory, a computer vision system is used to inspect products and identify defective items on a production line. The inspection system must accurately detect defects while processing products quickly enough to avoid slowing down production.

The system requirements are:

1. Accuracy must be 95% or above.
2. Processing time must not exceed 50 ms.
3. The available hardware supports moderate computational resources.

The available methods are compared below.

Comparison of Methods

Method	Accuracy	Processing Time	Computational Requirement	Evaluation
Traditional Image Analysis	91%	25 ms	Low	Fast but fails accuracy requirement
YOLO	96%	45 ms	Moderate	Meets all requirements
Deep Learning	98%	75 ms	High	Accurate but too slow

Evaluation of Traditional Image Analysis

Traditional Image Analysis has a processing time of 25 ms, making it the fastest method. It also requires low computational resources, which makes deployment simple and cost-effective.

However, its accuracy is only 91%, which is below the required minimum of 95%. This means that some defective products may not be detected and could continue through the production line.

Therefore:

- Accuracy requirement: $91\% < 95\%$ ✗

- Processing time: $25 \text{ ms} \leq 50 \text{ ms}$ ✓
- Computation: Low, supported by available hardware ✓

Although the method is fast and inexpensive, its low accuracy makes it unsuitable for quality-critical product inspection.

Evaluation of YOLO

YOLO achieves an accuracy of 96%, which is above the required minimum of 95%. Its processing time is 45 ms, which is below the maximum limit of 50 ms.

YOLO requires moderate computational resources, which matches the available hardware in the factory.

Therefore:

- Accuracy requirement: $96\% \geq 95\%$ ✓
- Processing time: $45 \text{ ms} \leq 50 \text{ ms}$ ✓
- Computation: Moderate = Available moderate resources ✓

YOLO can detect defects accurately while maintaining real-time performance. This helps the production line operate efficiently without significantly affecting speed.

Evaluation of Deep Learning

Deep Learning provides the highest accuracy of 98%, making it highly effective for detecting product defects.

However, its processing time is 75 ms, which exceeds the maximum acceptable limit of 50 ms. It also requires high computational resources, while the factory only supports moderate computational capability.

Therefore:

- Accuracy requirement: $98\% \geq 95\%$ ✓
- Processing time: $75 \text{ ms} > 50 \text{ ms}$ ✗
- Computation: High > Moderate available resources ✗

Although the accuracy is excellent, the system may slow down the production line and require additional expensive hardware.

Final Decision

YOLO is the best method for the smart factory product inspection system.

It is the only method that satisfies all the specified constraints:

- Accuracy is above 95%.
- Processing time is below 50 ms.
- Computational requirements match the available hardware.

Conclusion

Therefore, YOLO should be selected for the smart factory product inspection system. Traditional Image Analysis is faster but does not provide sufficient accuracy to reliably identify defects. Deep Learning provides the highest accuracy but requires more processing time and computational resources.

YOLO provides the best balance between high defect detection accuracy, fast processing, moderate computational requirements, and practical deployment cost. Hence, it is the most suitable solution for real-time industrial product inspection.

